MA 5104 Statistical Inference and Learning **due: 26-Aug-2026**
HW–2 Dept. of Mathematics, IIT Madras

Topic: Point and Confidence Estimation

Important note for submissions:

- Only handwritten answers are allowed.
- The hard-copy of solutions should be submitted to me in Wednesday lecture.
- No email submissions will be entertained.
- **Submit solutions to ONLY Problems 4, 5, 6, 7, and 10.**

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1. An efficiency expert wishes to determine the average time that it takes to drill three holes in a certain metal clamp. How large a sample will she need to be 95% confident that her sample mean will be within 15 seconds of the true mean? Assume that it is known from previous studies that $\sigma = 40$ seconds.

2. Regular consumption of presweetened cereals contributes to tooth decay, heart disease, and other degenerative diseases, according to studies conducted by Dr. W. H. Bowen of the National Institute of Health and Dr. J. Yudben, Professor of Nutrition and Dietetics at the University of London. In a random sample consisting of 20 similar single servings of Alpha-Bits, the average sugar content was 11.3 grams with a standard deviation of 2.45 grams. Assuming that the sugar contents are normally distributed, construct a 95% confidence interval for the mean sugar content for single servings of Alpha-Bits.

3. Construct a 95% confidence interval for σ^2 in Question 2 above.

4. The following data represent the running times of films produced by two motion-picture companies.

Company	**Time (minutes)**
I	103, 94, 110, 87, 98
II	97, 82, 123, 92, 175, 88, 118

Compute a 90% confidence interval for the difference between the average running times of films produced by the two companies. Assume that the running-time differences are approximately normally distributed with unequal variances.

5. Construct a 90% confidence interval for σ_1^2/σ_2^2 in Exercise 9.46 on page 295. Should we have assumed $\sigma_1^2 = \sigma_2^2$ in constructing our confidence interval for $\mu_I - \mu_{II}$?

6. Consider $S'^2 = \frac{1}{n}\sum(X_i - \overline{X})^2$, the estimator of σ^2.

   (a) What is the bias of S'^2?

   (b) Show that the bias of S'^2 approaches zero as $n \to \infty$.

7. A type of thread is being studied for its tensile strength properties. Fifty pieces were tested under similar conditions, and the results showed an average tensile strength of 78.3 kilograms and a standard deviation of 5.6 kilograms. Assuming a normal distribution of tensile strengths, give a

(a) 95% confidence interval on mean observed tensile strength value.

(b) lower 95% confidence limit on mean observed tensile strength value.

(c) upper 95% confidence limit on mean observed tensile strength value.

8. A manufacturer claims that at most 5% of its products are defective. In a random sample of 400 products, 28 are found to be defective.

   (a) Construct a 95% confidence interval for the true proportion of defective products.

   (b) Based on the interval, discuss whether the manufacturer's claim appears reasonable.

9. A survey finds that 46% of 800 randomly selected customers prefer Product A.

   A manager says:

   > "Since 46% prefer Product A, we can conclude that between 44% and 48% of all customers prefer Product A."

   Is this conclusion justified? Construct the appropriate 95% confidence interval and comment on the manager's statement.

10. Two students construct 95% confidence intervals for the same population proportion using two independent samples. Student A obtains $(0.42, 0.50)$, while Student B obtains $(0.38, 0.54)$.

    (a) Which student had the larger sample size? Can you determine this with certainty?

    (b) Which estimate is more precise?

    (c) What does the difference in widths tell you about the information contained in the two samples?